DSC 40A - Homework 7

Due: Wednesday, Dec 3th at 11:59PM

Homeworks are due to Gradescope by 11:59PM on the due date.

Note: Slip days cannot be used for Homework 7! We plan to release the solutions on Thursday so you have sufficient time to study for the final exam.

Homework will be evaluated not only on the correctness of your answers, but on your ability to present your ideas clearly and logically. You should **always explain and justify** your conclusions, using sound reasoning. Your goal should be to convince the reader of your assertions. If a question does not require explanation, it will be explicitly stated.

Homeworks should be written up and turned in by each student individually. You may talk to other students in the class about the problems and discuss solution strategies, but you should not share any written communication and you should not check answers with classmates. You can tell someone how to do a homework problem, but you cannot show them how to do it. **Only handwritten solutions will be accepted (use of tablets is permitted). Do not typeset your homework (using LaTeXor any other software)**.

For each problem you submit, you should **cite your sources** by including a list of names of other students with whom you discussed the problem. Instructors do not need to be cited.

Note: For full credit, make sure to assign pages to questions when you upload your submission to Gradescope. You will lose points if you don't!

Problem 1. Reflection and Feedback Form

a) Make sure to fill out this Reflection and Feedback Form, linked here, for two points on this homework! This form is primarily for your benefit; research shows that reflecting and summarizing knowledge helps you understand and remember it.

b) Fill out the Student Evaluation of Teaching (SET) for DSC40A. Feedback on the course is very helpful to us personally and the DSC program in general to understand what is working well in the course and what changes can be made to improve for the future. Your feedback helps drive the course development and impacts future students taking the course. This is especially important in a relatively new program such as ours.

Problem 2. Schrödi and the Boxes

A cat named **Schrödi** lives in a house with four identical boxes labeled **1**, **2**, **3**, and **4**. Each morning, Schrödi chooses one box uniformly at random to nap in. If Schrödi chooses to nap in Box 3, he always *meows* once.

a) Before you open any box, you wait and *do not* hear a meow. What is the probability that Schrödi is currently in box **4**?

Solution:

Let $S \in \{A, B, C, D\}$ be Schrödi's *final* box after any teleports.

Initially, Schrödi chooses a box uniformly, so

$$\mathbb{P}(S = 1) = \mathbb{P}(S = 2) = \mathbb{P}(S = 3) = \mathbb{P}(S = 4) = \frac{1}{4}.$$

On day 24, any sequence of 0–3 teleports leaves the final location uniform: the last position (either the initial choice or the last teleport) is always a uniform draw from $\{A, B, C, D\}$.

If we do *not* hear a meow, then $S \neq C$. Conditioning the uniform prior on $\{1, 2, 4\}$,

$$\mathbb{P}(S = 4 \mid \text{no meow}) = \frac{\frac{1}{4}}{\frac{3}{4}} = \boxed{\frac{1}{3}}.$$

b) You select a box uniformly at random and open it. The box is *empty*. You still have not heard any meow. Now what is the probability that Schrödi is currently in box **4**?

Solution:

Let O be the box you opened, chosen uniformly at random from $\{1, 2, 3, 4\}$, independent of S. Finding it empty means $O \neq S$, and "no meow" still implies $S \neq C$.

For $x \in \{1, 2, 4\}$,

$$\begin{aligned}
\mathbb{P}(S = x,\ O \neq S,\ \text{no meow}) &= \mathbb{P}(O \neq S \mid S = x, \text{no meow})\mathbb{P}(\text{no meow} \mid S = x)\mathbb{P}(S = x) && (1)\\
&= \underbrace{\frac{3}{4}}_{\mathbb{P}(O \neq S \mid S = x, \text{no meow})} \cdot \underbrace{1}_{\mathbb{P}(\text{no meow} \mid S = x)} \cdot \underbrace{\frac{1}{4}}_{\mathbb{P}(S = x)} = \frac{3}{16}. && (2)
\end{aligned}$$

(Alternative calculation:

$$\begin{aligned}
\mathbb{P}(S = x,\ O \neq S,\ \text{no meow}) &= \mathbb{P}(O \neq S \mid S = x, \text{no meow})\mathbb{P}(S = x \mid \text{no meow})\mathbb{P}(\text{no meow}) && (3)\\
&= \underbrace{\frac{3}{4}}_{\mathbb{P}(O \neq S \mid S = x, \text{no meow})} \cdot \underbrace{\frac{1}{3}}_{\mathbb{P}(\mid S = x \mid \text{no meow})} \cdot \underbrace{\frac{3}{4}}_{\mathbb{P}(\text{no meow})} = \frac{3}{16}. && (4)
\end{aligned}$$

)

For $x = 3$, this probability is 0 since "no meow" is impossible if $S = C$.

So

$$\mathbb{P}(S = D \mid O \neq S,\ \text{no meow}) = \frac{\frac{3}{16}}{3 \cdot \frac{3}{16}} = \boxed{\frac{1}{3}}.$$

c) Suppose you now open *two random boxes* (without replacement) and find that *both are empty*. You still have not heard any meow. What is the probability that Schrödi is in box **4**?

Solution:

Let O_1, O_2 be the two opened boxes. Both being empty means $S \notin \{O_1, O_2\}$, and "no meow" implies $S \neq 3$.

Fix $x \in \{1, 2, 4\}$. Then

$$\mathbb{P}(S = x,\ O_1, O_2 \text{ empty},\ \text{no meow}) = \underbrace{\frac{1}{4}}_{\mathbb{P}(S=x)} \cdot \underbrace{\frac{\binom{3}{2}}{\binom{4}{2}}}_{\text{choose 2 boxes from the 3 that don't contain } S} \cdot \underbrace{1}_{\text{no meow}} \tag{5}$$

$$= \frac{1}{4} \cdot \frac{3}{6} = \frac{1}{8}. \tag{6}$$

For $x = 3$, the probability is 0 (no meow impossible).

Normalizing over $S \in \{1, 2, 4\}$,

$$\mathbb{P}(S = D \mid O_1, O_2 \text{ empty},\ \text{no meow}) = \frac{\frac{1}{8}}{3 \cdot \frac{1}{8}} = \boxed{\frac{1}{3}}.$$

d) Suppose instead that you *did* hear a meow, and then opened one random box and found it empty. What is the probability that Schrödi is in box **3**?

Solution:

Schrödi meows *if and only if* his final location is box 3. So hearing a meow already tells us $S = 3$ with certainty:

$$\mathbb{P}(S = 3 \mid \text{meow heard}) = 1.$$

Opening a random box and finding it empty just tells us that the opened box $\neq 3$, which is consistent with $S = 3$ and does not change the probability.

Therefore,

$$\mathbb{P}(S = C \mid \text{meow heard, one empty box}) = \boxed{1}.$$

Problem 3. Independence and Complements

a) Let A and B be two independent events in the sample space S. Show that $\bar{A}$ and $\bar{B}$ must be independent of one another. Make sure your proof still works in the special case that one or more of A, B, $\bar{A}$, $\bar{B}$ has probability zero. **Hint:** It helps to draw a Venn diagram.

Solution: Let A, B be independent so $P(A \cap B) = P(A) \cdot P(B)$. To show that $\bar{A}$ and $\bar{B}$ are independent, we must show that $P(\bar{A} \cap \bar{B}) = P(\bar{A}) \cdot P(\bar{B})$. The first thing to notice is that $P(\bar{A} \cap \bar{B}) = 1 - P(A \cup B)$. This can be demonstrated via a Venn diagram.

Using this fact, we have

$$P(\bar{A} \cap \bar{B}) = 1 - P(A \cup B).$$

By the addition rule,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Therefore, since $P(A \cap B) = P(A)(B)$,

$$\begin{aligned} P(\bar{A} \cap \bar{B}) &= 1 - \big(P(A) + P(B) - P(A \cap B)\big) \\ &= 1 - P(A) - P(B) + P(A)(B) \\ &= (1 - P(A)) \cdot (1 - P(b)) \\ &= P(\bar{A})(\bar{B}). \end{aligned}$$

This shows that when A and B are independent, so are their complements.

b) Let E and F be two events in a sample space S, with $0 < \mathbb{P}(F) < 1$. If $\mathbb{P}(E|F) = \mathbb{P}(E|\overline{F})$, must it be true that E and F are independent? Provide a proof of independence, or give a counterexample by specifying a sample space S and two dependent events E and F that satisfy the given conditions.

Solution: Yes, it must be true that E and F are independent in this case.

Since $0 < \mathbb{P}(F) < 1$, we know that $\mathbb{P}(F) \neq 0$ and $\mathbb{P}(\overline{F}) \neq 0$. Thus, we can divide by both $\mathbb{P}(F)$ and $\mathbb{P}(\overline{F})$. As a result, each of the following statements is equivalent.

$$\begin{aligned} \mathbb{P}(E|F) &= \mathbb{P}(E|\overline{F}) \\ \frac{\mathbb{P}(E \cap F)}{\mathbb{P}(F)} &= \frac{\mathbb{P}(E \cap \overline{F})}{\mathbb{P}(\overline{F})} \\ \frac{\mathbb{P}(E \cap F)}{\mathbb{P}(F)} &= \frac{\mathbb{P}(E \cap \overline{F})}{1 - \mathbb{P}(F)} \\ \mathbb{P}(E \cap F) \cdot (1 - \mathbb{P}(F)) &= \mathbb{P}(E \cap \overline{F}) \cdot \mathbb{P}(F) \\ \mathbb{P}(E \cap F) - \mathbb{P}(E \cap F) \cdot \mathbb{P}(F) &= \mathbb{P}(E \cap \overline{F}) \cdot \mathbb{P}(F) \\ \mathbb{P}(E \cap F) &= \mathbb{P}(E \cap F) \cdot \mathbb{P}(F) + \mathbb{P}(E \cap \overline{F}) \cdot \mathbb{P}(F) \\ \mathbb{P}(E \cap F) &= (\mathbb{P}(E \cap F) + \mathbb{P}(E \cap \overline{F})) \cdot \mathbb{P}(F) \\ \mathbb{P}(E \cap F) &= \mathbb{P}(E) \cdot \mathbb{P}(F) \end{aligned}$$

This is exactly what it means for E and F to be independent.

The last step is a consequence of the Law of Total Probability. Since F and $\overline{F}$ partition the sample space, $\mathbb{P}(E \cap F) + \mathbb{P}(E \cap \overline{F}) = \mathbb{P}(E)$.

An even cleaner solution uses the Law of Total Probability in the first step:

$$\begin{aligned}
\mathbb{P}(E) &= \mathbb{P}(E \cap F) + \mathbb{P}(E \cap \overline{F}) \\
&= \mathbb{P}(E|F) \cdot \mathbb{P}(F) + \mathbb{P}(E|\overline{F}) \cdot \mathbb{P}(\overline{F}) \\
&= \mathbb{P}(E|F) \cdot \mathbb{P}(F) + \mathbb{P}(E|F) \cdot \mathbb{P}(\overline{F}) \\
&= \mathbb{P}(E|F) \left(\mathbb{P}(F) + \mathbb{P}(\overline{F})\right) \\
&= \mathbb{P}(E|F) \cdot 1 \\
&= \mathbb{P}(E|F)
\end{aligned}$$

Since $\mathbb{P}(E|F) = \mathbb{P}(E)$, E and F are independent.

c) Let E and F be two events in a sample space S, with $0 < \mathbb{P}(F) < 1$. If $\mathbb{P}(E|F) = \mathbb{P}(\overline{E}|F)$, must it be true that E and F are independent? Provide a proof of independence, or give a counterexample by specifying a sample space S and two dependent events E and F that satisfy the given conditions.

Solution: No, it need not be the case that E and F are independent.

For example, consider the sample space $S = \{1, 2, 3, 4, 5, 6\}$ where each outcome is equally likely (the results of a die roll, for example).

Let $E = \{1\}$ and $F = \{1, 2\}$. Then $\mathbb{P}(E|F) = \frac{1}{2}$. Then $\overline{E} = \{2, 3, 4, 5, 6\}$, so $\mathbb{P}(\overline{E}|F) = \frac{1}{2}$. Therefore, $\mathbb{P}(E|F) = \mathbb{P}(\overline{E}|F)$ as required.

However this E and F are dependent because $\mathbb{P}(E) = \frac{1}{6}$ and $\mathbb{P}(E|F) = \frac{1}{2}$ so $\mathbb{P}(E) \neq \mathbb{P}(E|F)$.

Problem 4. Genetic test

Let's revisit the example we studied in class. 1% of the population have a certain genetic defect.

a) A test has been developed such that 90% of administered tests accurately detect the gene (true positives). What needs to be the false positive probability so that the probability of a patient having the genetic defect given a positive result (the posterior) is 90%?

Solution:

- Hypothesis: patient has the gene, $P(H) = 0.01$
- Evidence: patient got a positive test result, $P(E)$
- True positive: Probability of positive test result if someone has the gene $P(E|H) = 0.9$
- False positive: Probability of positive test result if someone doesn't have the gene, denote $P(E|\bar{H}) = p$

$$\begin{aligned} P(H|E) &= \frac{P(E|H)P(H)}{P(E)} \\ &= \frac{P(E|H)P(H)}{P(E|H)P(H) + P(E|\bar{H})P(\bar{H})} \\ &= \frac{0.9 \cdot 0.01}{0.9 \cdot 0.01 + p \cdot 0.99} = 0.9 \end{aligned}$$

Rearranging the terms

$$\begin{aligned} 0.9 \cdot 0.01 &= 0.9 \cdot (0.9 \cdot 0.01 + p \cdot 0.99) \\ 0.1 \cdot 0.9 \cdot 0.01 &= 0.9 \cdot p \cdot 0.99 \\ p = P(E|\bar{H}) &= \frac{0.1 \cdot 0.01}{0.99} \approx 0.001 \end{aligned}$$

The probability of a false positive (positive result for someone who doesn't have the gene) needs to be 0.1%.

For subproblems (b)-(d), consider the following. A test has been developed such that 1% of administered tests are positive when the patient doesn't have the gene (false positives).

b) What needs to be the true positive probability so that the probability of a patient having the genetic defect given a positive result (the posterior) is 50%?

Solution:

- Hypothesis: patient has the gene, $P(H) = 0.01$
- True positive: Probability of positive test result if someone has the gene, denote $P(E|H) = p$
- False positive: Probability of positive test result if someone doesn't have the gene, denote $P(E|\bar{H}) = 0.01$

$$
\begin{aligned}
P(H|E) &= \frac{P(E|H)P(H)}{P(E|H)P(H) + P(E|\bar{H})P(\bar{H})} \\
&= \frac{p \cdot 0.01}{p \cdot 0.01 + 0.01 \cdot 0.99} \\
&= \frac{p}{p + 0.99} = 0.5
\end{aligned}
$$

Rearranging the terms

$$
\begin{aligned}
p &= 0.5 \cdot (p + 0.99) \\
0.5 \cdot p &= 0.5 \cdot 0.99 \\
p = P(E|H) &= 0.99
\end{aligned}
$$

The probability of a true positive needs to be 0.99%.

c) Show that there is no true positive probability such that the probability of a patient having the genetic defect given a positive result (the posterior) can be 90%. (Hint: remember a probability p needs to fulfill $0 \leq p \leq 1$.)

Solution:

- Hypothesis: patient has the gene, $P(H) = 0.01$
- True positive: Probability of positive test result if someone has the gene, denote $P(E|H) = p$
- False positive: Probability of positive test result if someone doesn't have the gene, denote $P(E|\bar{H}) = 0.01$

Let's try solving as before:

$$
\begin{aligned}
P(H|E) &= \frac{P(E|H)P(H)}{P(E|H)P(H) + P(E|\bar{H})P(\bar{H})} \\
&= \frac{p \cdot 0.01}{p \cdot 0.01 + 0.01 \cdot 0.99} \\
&= \frac{p}{p + 0.99} = 0.9
\end{aligned}
$$

Rearranging the terms

$$
\begin{aligned}
p &= 0.9 \cdot (p + 0.99) \\
0.1 \cdot p &= 0.9 \cdot 0.99 \\
p = P(E|H) &= \frac{0.99 \cdot 0.9}{0.1} = 8.91 > 1
\end{aligned}
$$

Given the rate of false positives, we cannot find a true positive rate such that the posterior is 0.9.

d) What is the highest *posterior* probability such a test can have? What is the corresponding true positive probability? (Hint: remember a probability p needs to fulfill $0 \leq p \leq 1$.)

Solution:

- Hypothesis: patient has the gene, $P(H) = 0.01$
- Evidence: patient got a positive test result, $P(E)$
- True positive: Probability of positive test result if someone has the gene, denote $P(E|H) = p$
- False positive: Probability of positive test result if someone doesn't have the gene, denote $P(E|\bar{H}) = 0.01$

Following our previous calculations

$$\begin{aligned} P(H|E) &= \frac{p \cdot 0.01}{p \cdot 0.01 + 0.01 \cdot 0.99} \\ &= \frac{p}{p + \cdot 0.99} \end{aligned}$$

We will denote the posterior probability $P(H|E) = q$.

Rearranging the terms

$$\begin{aligned} p &= q \cdot (p + 0.99) \\ (1 - q) \cdot p &= q \cdot 0.99 \\ p &= \frac{0.99 \cdot q}{1 - q} \end{aligned}$$

By definition $p \leq 1$, therefore

$$\begin{aligned} p = \frac{0.99 \cdot q}{1 - q} &\leq 1 \\ 0.99 * q &\leq 1 - q \\ q &\leq \frac{1}{1.99} \approx 0.502 \end{aligned}$$

The maximum posterior value is $\approx 50.2\%$.

The true positive probability is then

$$p = \frac{0.99 \cdot q}{1 - q} = \frac{0.99 \cdot \frac{1}{1.99}}{1 - \frac{1}{1.99}} = 1$$

Alternative solution:

Some students used $p = P(E|H) = 0.9$ given in (a). In this case:

$$\begin{aligned} P(H|E) &= \frac{p}{p + \cdot 0.99} \\ &= \frac{0.9}{0.9 + \cdot 0.99} \approx 0.48 \end{aligned}$$

Problem 5. Avi's music

Avi has a very unique music taste. Whenever he listens to a particular song, he would only enjoy it sometimes, depending on the artist of that song. Avi enjoyed songs by:

- Kendrick Lamar 90% of the time,
- Taylor Swift 75% of the time,
- Drake 45% of the time, and
- J Cole 50% of the time.

a) Avi played a song from one of the above four artists and he really enjoyed the song. You have no idea which of the four artists he listened to, so assume it was equally likely to be any of them. You can also assume that Avi only listens to those 4 artists.

Given that Avi enjoyed the song, what's the probability that it was a song from Kendrick Lamar? Taylor Swift? Drake? J Cole? Show your work.

Solution: Throughout this question, we'll define the following events:

- E: He enjoys the song.
- K: The song is by Kendrick Lamar.
- T: The song is by Taylor Swift.
- D: The song is by Drake.
- J: The song is by J Cole.

We're given the following information:

- The artist is equally likely to be any of them, so $\mathbb{P}(K) = \mathbb{P}(T) = \mathbb{P}(D) = \mathbb{P}(J) = 0.25$.
- $\mathbb{P}(E|K) = 0.9$, $\mathbb{P}(E|T) = 0.75$, $\mathbb{P}(E|D) = 0.45$, and $\mathbb{P}(E|J) = 0.4$.

We need to find four probabilities, $\mathbb{P}(K|E)$, $\mathbb{P}(T|E)$, $\mathbb{P}(D|E)$, and $\mathbb{P}(J|E)$. To do so, we will use Bayes rule in each case, which states that, for any artist A,

$$\mathbb{P}(A|E) = \frac{\mathbb{P}(A)\mathbb{P}(E|A)}{\mathbb{P}(E)}$$

Note that we will use this same formula for all four probabilities, and all four probabilities have $\mathbb{P}(E)$ as part of their calculation. Furthermore, since the events K, T, D, and J form a partition of our sample space (exactly one of these events must be true), we can use the law of total probability to re-write $\mathbb{P}(E)$ in terms of the information we're given:

$$\begin{aligned}
\mathbb{P}(E) &= \mathbb{P}(E \cap K) + \mathbb{P}(E \cap T) + \mathbb{P}(E \cap D) + \mathbb{P}(E \cap J) \\
\mathbb{P}(E) &= \mathbb{P}(K)\mathbb{P}(E|K) + \mathbb{P}(T)\mathbb{P}(E|T) + \mathbb{P}(D)\mathbb{P}(E|D) + \mathbb{P}(J)\mathbb{P}(E|J) \\
&= 0.25 \cdot 0.9 + 0.25 \cdot 0.75 + 0.25 \cdot 0.45 + 0.25 \cdot 0.5 \\
&= 0.65
\end{aligned}$$

Now, we can compute the required probabilities as follows:

$$
\begin{aligned}
\mathbb{P}(K|E) &= \frac{\mathbb{P}(K)\mathbb{P}(E|K)}{\mathbb{P}(E)} = \frac{0.25 \cdot 0.9}{0.65} \approx 0.346 \\
\mathbb{P}(T|E) &= \frac{\mathbb{P}(C)\mathbb{P}(E|T)}{\mathbb{P}(E)} = \frac{0.25 \cdot 0.75}{0.65} \approx 0.288 \\
\mathbb{P}(D|E) &= \frac{\mathbb{P}(D)\mathbb{P}(E|D)}{\mathbb{P}(E)} = \frac{0.25 \cdot 0.45}{0.65} \approx 0.173 \\
\mathbb{P}(J|E) &= \frac{\mathbb{P}(J)\mathbb{P}(E|J)}{\mathbb{P}(E)} = \frac{0.25 \cdot 0.5}{0.65} \approx 0.192
\end{aligned}
$$

As expected, the artist is most likely to be Kendrick Lamar, since that's the artist that Avi likes best.

b) Avi again listens to a song from one of the above 4 artists and enjoys the song. This time, instead of assuming that he's equally likely to listen to all four artists, suppose you know that Avi listens to:

- Kendrick Lamar 20% of the time,
- Taylor Swift 35% of the time,
- Drake 15% of the time, and
- J Cole 30% of the time.

Given that Avi enjoyed the song, what's the probability that the song was by Kendrick Lamar? Taylor Swift? Drake? J Cole? Show your work.

Solution:

We can use the same framework as in the last part, it's just that $\mathbb{P}(K)$, $\mathbb{P}(T)$, $\mathbb{P}(D)$, and $\mathbb{P}(J)$ have all changed. Specifically, $\mathbb{P}(K) = 0.15$, $\mathbb{P}(T) = 0.4$, $\mathbb{P}(D) = 0.1$, and $\mathbb{P}(J) = 0.35$.

$$
\begin{aligned}
\mathbb{P}(E) &= \mathbb{P}(E \cap K) + \mathbb{P}(E \cap T) + \mathbb{P}(E \cap D) + \mathbb{P}(E \cap J) \\
\mathbb{P}(E) &= \mathbb{P}(K)\mathbb{P}(E|K) + \mathbb{P}(T)\mathbb{P}(E|T) + \mathbb{P}(D)\mathbb{P}(E|D) + \mathbb{P}(J)\mathbb{P}(E|J) \\
&= 0.2 \cdot 0.9 + 0.35 \cdot 0.75 + 0.15 \cdot 0.45 + 0.3 \cdot 0.5 \\
&= 0.66
\end{aligned}
$$

Then:

$$\mathbb{P}(K|E) = \frac{\mathbb{P}(K)\mathbb{P}(E|K)}{\mathbb{P}(E)} = \frac{0.2 \cdot 0.9}{0.66} \approx 0.273$$
$$\mathbb{P}(T|E) = \frac{\mathbb{P}(T)\mathbb{P}(E|T)}{\mathbb{P}(E)} = \frac{0.35 \cdot 0.75}{0.66} \approx 0.398$$
$$\mathbb{P}(D|E) = \frac{\mathbb{P}(D)\mathbb{P}(E|D)}{\mathbb{P}(E)} = \frac{0.15 \cdot 0.45}{0.66} \approx 0.102$$
$$\mathbb{P}(J|E) = \frac{\mathbb{P}(J)\mathbb{P}(E|J)}{\mathbb{P}(E)} = \frac{0.3 \cdot 0.5}{0.66} \approx 0.227$$

Now it's most likely that the song was by Taylor Swift, which makes sense because Taylor Swift is his most listened to artist, and she is also an artist that he enjoys.

c) Compare your answers to part (a) and part (b) above. Identify which of the four probabilities you computed increased and which decreased, and explain why this makes sense intuitively.

Solution: The probability that the artist was Kendrick Lamar or Drake has decreased, which makes sense because Avi was less likely to listen to those artists.

The probability that the artist was Taylor Swift or J Cole increased, which makes sense since we know Avi listens to those artists a lot.

Notice how our prior knowledge about the frequency of Avi listening to an artist changes our results. Having this knowledge leads to different results than in part (a), when we did not have this knowledge.

Problem 6. Oops, you lost the King of Clubs!

You now have a 51-card deck obtained from a standard 52-card deck by removing the King of Clubs, denoted $K\clubsuit$. You draw one card at random from this deck.

To visualize the deck, here are the cards organized by suit:

$$\begin{aligned}
\heartsuit &: \ 2,3,4,5,6,7,8,9,10,J,Q,K,A \\
\diamondsuit &: \ 2,3,4,5,6,7,8,9,10,J,Q,K,A \\
\clubsuit &: \ 2,3,4,5,6,7,8,9,10,J,Q,__,A \\
\spadesuit &: \ 2,3,4,5,6,7,8,9,10,J,Q,K,A
\end{aligned}$$

- Let A be the event that the card is a heart.
- Let B be the event that the card is a face card (i.e., J, Q, K).
- Let C be the event that the card is *red* (i.e., hearts or diamonds).

a) Are A and B independent? Justify your answer.

Solution: There are 51 cards total.

There are 13 hearts, so

$$\mathbb{P}(A) = \frac{13}{51}.$$

Face cards are J, Q, K. In this deck there are

$$3 \text{ hearts} + 3 \text{ diamonds} + 3 \text{ spades} + 2 \text{ clubs} = 11$$

face cards, so

$$\mathbb{P}(B) = \frac{11}{51}.$$

The event $A \cap B$ is "the card is a heart and a face card", so the possible cards are

$$J\heartsuit,\ Q\heartsuit,\ K\heartsuit,$$

three cards in total. Hence

$$\mathbb{P}(A \cap B) = \frac{3}{51}.$$

If A and B were independent, we would have

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B) = \frac{13}{51} \cdot \frac{11}{51} = \frac{143}{2601},$$

but

$$\frac{3}{51} = \frac{153}{2601} \neq \frac{143}{2601}.$$

Therefore A and B are *not* independent.

(Equivalently, $\mathbb{P}(B \mid A) = 3/13 \neq 11/51 = \mathbb{P}(B)$.)

b) Now suppose you learn that the card is red, i.e., that C occurred. Are A and B independent given this new information? In other words, are A and B conditionally independent given C? Justify your answer.

Solution: Conditioning on C restricts the sample space to the 26 red cards (13 hearts and 13 diamonds).

Among these red cards,

$$\mathbb{P}(A \mid C) = \frac{\#\text{ hearts}}{\#\text{ red cards}} = \frac{13}{26} = \frac{1}{2}.$$

The face cards among the red cards are

$$J\heartsuit, Q\heartsuit, K\heartsuit, \quad J\diamondsuit, Q\diamondsuit, K\diamondsuit,$$

so there are 6 of them, and

$$\mathbb{P}(B \mid C) = \frac{6}{26}.$$

The event $A \cap B \cap C$ is "the card is a red heart and a face card", so the possible cards are

$$J\heartsuit, Q\heartsuit, K\heartsuit,$$

three cards. Thus

$$\mathbb{P}(A \cap B \mid C) = \frac{3}{26}.$$

Now check conditional independence:

$$\mathbb{P}(A \mid C)\mathbb{P}(B \mid C) = \frac{1}{2} \cdot \frac{6}{26} = \frac{6}{52} = \frac{3}{26} = \mathbb{P}(A \cap B \mid C).$$

Since

$$\mathbb{P}(A \cap B \mid C) = \mathbb{P}(A \mid C)\mathbb{P}(B \mid C),$$

the events A and B *are* independent given C.

Problem 7. Comic Characters

In this problem, we'll work with a dataset of characters from comic books. At first, we look at a limited set of 25 of the more popular comic characters. You can find this dataset on the last page of this assignment as well as on Datahub in the same directory as the supplementary Jupyter notebook (linked). Note that, the data (csv files) used are linked here.

For each character, we have the following information.

Variable	Definition
`ID`	The identity status of the character ("Public Identity" or "Secret Identity")
`SEX`	Sex of the character ("Male Characters" or "Female Characters")
`ALIVE`	Whether the character is alive ("Living Characters" or "Deceased Characters")
`COMPANY`	The comic company that created the character ("DC" or "Marvel")
`ALIGN`	The alignment of the character ("Bad Characters", "Good Characters", or "Neutral Characters")

a) Use the set of 25 popular characters to make a prediction using Naive Bayes (without smoothing) for the following character's alignment:

- "Secret Identity"
- "Male Characters"
- "Living Characters"
- "Marvel"

Your prediction should be "Bad Characters", "Good Characters", or "Neutral Characters", whichever is most likely according to Naive Bayes. Do this part by hand, and show your work. You can check your answer using the code you'll write in part (b).

Solution:

When performing classification with naive Bayes, we wish to find the class label that maximizes the following expression:

$$P(\text{class}|\text{features}) = \frac{P(\text{features}|\text{class}) \cdot P(\text{class})}{P(\text{features})}$$

Note that since $P(\text{features})$ is constant (features are given in the question), we can omit the denominator, and just try to maximize:

$$P(\text{features}|\text{class}) \cdot P(\text{class})$$

The class that maximizes this quantity will be our prediction.

In this problem, there are three possible classes, "Bad Characters", "Good Characters", or "Neutral Characters"

Class 1. "Bad Characters":

$P(\text{Bad}) = 2/25$

$P((\text{Secret, Male, Living, Marvel})|\text{Bad}) = 1/2 * 2/2 * 2/2 * 1/2$ according to the conditional independence assumption.

Overall, $P((\text{Secret, Male, Living, Marvel})|\text{Bad}) \cdot P(\text{Bad}) = 1/2 * 2/2 * 2/2 * 1/2 * 2/25 = \mathbf{0.02}$

Class 2. "Good Characters":

$P(\text{Good}) = 19/25$

$P((\text{Secret, Male, Living, Marvel})|\text{Good}) = 7/19*16/19*16/19*13/19$ according to the conditional independence assumption.

Overall, $P((\text{Secret, Male, Living, Marvel})|\text{Good}) \cdot P(\text{Good}) = 7/19*16/19*16/19*13/19*19/25 \approx$ **0.136**

Class 3. "Neutral Characters":

$P(\text{Neutral}) = 4/25$

$P((\text{Secret, Male, Living, Marvel})|\text{Neutral}) = 2/4 * 4/4 * 3/4 * 4/4$ according to the conditional independence assumption.

Overall, $P((\text{Secret, Male, Living, Marvel})|\text{Neutral}) \cdot P(\text{Neutral}) = 2/4*4/4*3/4*4/4*4/25 =$ **0.06**

b) 🥑🥑🥑🥑 In the supplementary Jupyter notebook (linked), we've provided not only the dataset of 25 popular characters, but also a much larger dataset of over ten thousand comic characters. For this part, you will be implementing a Naive Bayes classifier (without smoothing) to predict the alignment of any character based on their features (ID, SEX, ALIVE, COMPANY), using the larger dataset.

Complete the function `predict_align`, which takes in a DataFrame of comic characters and the features of one particular character, and returns the predicted alignment for that character according to Naive Bayes without smoothing, using the input DataFrame of characters.

Your function should return a string, either "Bad Characters", "Good Characters", or "Neutral Characters".

Using your code, predict the alignment of Spiderman. You do not need to submit your code.

Solution: One solution is given below:

```python
def predict_align(df, identity, sex, alive, company):
    '''Takes in a DataFrame of comic characters and the features of one particular character,
    and returns the predicted alignment for that character according to Naive Bayes without smoothing,
    using the input DataFrame of characters.'''

    results = np.array([])
    for align in ["Bad Characters", "Good Characters", "Neutral Characters"]:

        this_align = df[df.get("ALIGN") == align]
        num_this_align = this_align.shape[0]
        p_this_align = num_this_align/df.shape[0]

        id_term = this_align[this_align.get("ID") == identity].shape[0]/num_this_align
        sex_term = this_align[this_align.get("SEX") == sex].shape[0]/num_this_align
        alive_term = this_align[this_align.get("ALIVE") == alive].shape[0]/num_this_align
        company_term = this_align[this_align.get("COMPANY") == company].shape[0]/num_this_align

        result = p_this_align*id_term*sex_term*alive_term*company_term
        print(align, p_this_align, id_term, sex_term, alive_term, company_term, result)
        results = np.append(results, result)

    return ["Bad Characters", "Good Characters", "Neutral Characters"][np.argmax(results)]
```

c) 🥑🥑🥑🥑 Use your `predict_align` function to predict the alignment for the same character as you did in part (a), this time using the larger dataset of comic characters. As a reminder, the features were

- "Secret Identity"

- “Male Characters”
- “Living Characters”
- “Marvel”

You should get a different prediction than you got in part (a), when you used only the 25 popular characters. Why do you get different results? What specific difference between the two datasets explains why your predictions are different?

Hint: If you're stuck, try printing out each term in the products that you calculate in `predict_align`. Compare these terms when you input the DataFrame of 25 popular characters and when you input the larger DataFrame.

Hint: Another way to get started on this is to try other combinations of features as input to `predict_align`. When using the DataFrame of 25 popular characters, try to find a combination of features that leads to a different result than you got in part (a).

Solution: The difference between the two datasets that explains the different predictions is the balance betweeen the classes. In the smaller dataset of popular characters, 19 out of 25, or 76% of characters are good. Only 8% are bad and 16% are neutral. This gives a major advantage to the prediction “Good Characters”, because one of the terms in the Naive Bayes calculation is substantially higher for that class than for the other classes.

In fact, for a dataset with classes this imbalanced, the Naive Bayes classifier will always predict “Good Characters”, regardless of the features that it receives as input.

```
for identity in ["Public Identity", "Secret Identity"]:
    for sex in  ["Male Characters", "Female Characters"]:
        for alive in ["Living Characters", "Deceased Characters"]:
            for company in ["DC", "Marvel"]:
                print(predict_align(popular, identity, sex, alive, company))
```

```
Good Characters
Good Characters
Good Characters
Good Characters
Good Characters
Good Characters
Good Characters
Good Characters
Good Characters
Good Characters
Good Characters
Good Characters
Good Characters
Good Characters
Good Characters
Good Characters
```

On the other hand, in the larger dataset, the classes are more well-balanced. As we can see below, about 47% of characters are bad, 39% of characters are good, and 14% are neutral.

```
grouped = comics.groupby('ALIGN').count()
grouped
```

	ID	SEX	ALIVE	COMPANY
ALIGN				
Bad Characters	6433	6433	6433	6433
Good Characters	5316	5316	5316	5316
Neutral Characters	1845	1845	1845	1845

```
grouped.get('ID')/grouped.get('ID').sum()*100
```

```
ALIGN
Bad Characters        47.322348
Good Characters       39.105488
Neutral Characters    13.572164
Name: ID, dtype: float64
```

This means the playing field is more level, and in fact there are certain combinations of features that lead to predictions of "Bad Characters" and others that lead to predictions of "Good Characters." None lead to a prediction of "Neutral Characters", which makes sense because neutral characters are pretty rare.

name	ID	SEX	ALIVE	COMPANY	ALIGN
Wilson Fisk (Earth-616)	Public Identity	Male Characters	Living Characters	Marvel	Bad Characters
Joker (New Earth)	Secret Identity	Male Characters	Living Characters	DC	Bad Characters
Kent Nelson (New Earth)	Secret Identity	Male Characters	Deceased Characters	DC	Good Characters
Timothy Dugan (Earth-616)	Public Identity	Male Characters	Deceased Characters	Marvel	Good Characters
Samuel Wilson (Earth-616)	Public Identity	Male Characters	Living Characters	Marvel	Good Characters
Hulk (Robert Bruce Banner)	Public Identity	Male Characters	Living Characters	Marvel	Good Characters
Reed Richards (Earth-616)	Public Identity	Male Characters	Living Characters	Marvel	Good Characters
Benjamin Grimm (Earth-616)	Public Identity	Male Characters	Living Characters	Marvel	Good Characters
Iron Man (Anthony \"Tony\" Stark)	Public Identity	Male Characters	Living Characters	Marvel	Good Characters
Jessica Drew (Earth-616)	Secret Identity	Female Characters	Living Characters	Marvel	Good Characters
Johnathon Blaze (Earth-616)	Secret Identity	Male Characters	Living Characters	Marvel	Good Characters
Brian Braddock (Earth-616)	Public Identity	Male Characters	Living Characters	Marvel	Good Characters
Dane Whitman (Earth-616)	Secret Identity	Male Characters	Living Characters	Marvel	Good Characters
Captain America (Steven Rogers)	Public Identity	Male Characters	Living Characters	Marvel	Good Characters
Crystalia Amaquelin (Earth-616)	Public Identity	Female Characters	Living Characters	Marvel	Good Characters
Superman (Clark Kent)	Secret Identity	Male Characters	Living Characters	DC	Good Characters
Batman (Bruce Wayne)	Secret Identity	Male Characters	Living Characters	DC	Good Characters
Franklin Rock (New Earth)	Public Identity	Male Characters	Living Characters	DC	Good Characters
Cassandra Sandsmark (New Earth)	Public Identity	Female Characters	Living Characters	DC	Good Characters
Garth (New Earth)	Public Identity	Male Characters	Deceased Characters	DC	Good Characters
James Rhodes (Earth-616)	Secret Identity	Male Characters	Living Characters	Marvel	Good Characters
Deadpool (Wade Wilson)	Secret Identity	Male Characters	Living Characters	Marvel	Neutral Characters
Odin Borson (Earth-616)	Public Identity	Male Characters	Living Characters	Marvel	Neutral Characters
Otto Octavius (Earth-616)	Secret Identity	Male Characters	Deceased Characters	Marvel	Neutral Characters
Wolverine (James \"Logan\" Howlett)	Public Identity	Male Characters	Living Characters	Marvel	Neutral Characters